Chapter 2: Quantum Channels

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1. Basics of channels

Channels" describe the discrete evolution in time of a system containing quantum information. This includes useful operations and changes that must be avoided (noise).Common names for generic channels include Φ, Ψ, Ξ.

If a Φ channel is applied to a system in a state represented by a ρ density matrix, then we obtain a system in the $\Phi(\rho)$ state.

Definition:

Requirements:

> 1. Channels are linear mapping
> 2. Channels transform density matrices into density matrices.(1)

(1) this includes the possibility that a channel is applied as part of a compound system.

- Channels are linear mapping :

Suppose we have two density matrices ρ, σ, which pass through a channel Φ, with probability p and $1-p$ respectively. In this case, we have:

All Φ channels have an input X and an output Y, i.e. Φ transforms X into Y. It's possible that X and Y can be identical.

- Channels transform density matrices into density matrices.

Suppose Z is an additional system, Γ is a set of Z's class states, and the pair (Z, X)

- is in the state ρ. We can express ρ in the following form:

$$\rho = \sum_{a,b \in \Gamma} |a\rangle\langle b| \otimes \rho_{a,b}$$

Applied to the Φ channel which transforms X into Y, leaving (Z, Y) in the state:

$$\sum_{a,b \in \Gamma} |a\rangle\langle b| \otimes \Phi(\rho_{a,b})$$

we obtain a block matrix, where Φ is applied to each $\rho_{i,j}$ of the ρ density matrix.

Unitary Channels:
Suppose U is a unit matrix representing the unit operation on a system X.

The Φ channel corresponding to this operation is defined as follows:

$$\Phi(\rho) = U\rho U^\dagger$$

This can be verified by applying U to a ψ ket.Example with identity such as identity channel:

$$Id(\rho) = \rho$$

Convex combinations:
Suppose Φ_0, Φ_1 are channels from X to Y with $p \in [0, 1]$.

Suppose we apply Φ_0 with probability p and Φ_1 with probab $1 - p$. We obtain the new channel:

$$\Psi = p\Phi_O + (1 - p)\Phi_1$$

More generally, for a set of sub-channels with respective probabilities:

$$\Psi = \sum_{k=0}^{m-1} p_k \Phi_k$$

Example of unitary channels:
<u>Mixed unitary channels</u>
Applying a collection of unitary operations to a system results in a mixed unitary channel

$$\Psi(\rho) = \sum_{k=0}^{m-1} p_k U_k \rho U_k^\dagger$$

Pauli channels are channels whose unit matrices are all Pauli matrices.

<u>Qubit Reset Channel</u>:
Reset a qubit to state 0

$$\Lambda(\rho) = Tr(\rho)|0\rangle\langle 0|$$

Suppose A and B are qubits (A, B) is in a $|\phi^+\rangle$ state and Λ is applied to A. Both states A, B are in a Bell state $|\phi^+\rangle$, so we obtain the following density matrix:

$$|\phi^+\rangle\langle\phi^+| = \begin{pmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}$$

Using Dirac notation we have alternatively this result:

$$|\phi^+\rangle\langle\phi^+| = \frac{1}{2}|0\rangle\langle 0| \otimes |0\rangle\langle 0| + \frac{1}{2}|0\rangle\langle 1| \otimes |0\rangle\langle 1| + \frac{1}{2}|1\rangle\langle 0| \otimes |1\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| \otimes$$

Applying the channel effect reset 0 to A and doing nothing to B, we obtain:

$$\frac{1}{2}\Lambda(|0\rangle\langle 0| \otimes |0\rangle\langle 0|) + \Lambda(\frac{1}{2}|0\rangle\langle 1| \otimes |0\rangle\langle 1|) + \Lambda(\frac{1}{2}|1\rangle\langle 0| \otimes |1\rangle\langle 0|) + \Lambda(\frac{1}{2}|1\rangle\langle 1| \otimes$$

$$= \frac{1}{2}\Lambda(|0\rangle\langle 0| \otimes |0\rangle\langle 0|) + \frac{1}{2}\Lambda(|0\rangle\langle 0| \otimes |1\rangle\langle 1|) \ = \frac{1}{2}\Lambda(|0\rangle\langle 0| \otimes \frac{\mathbb{I}}{2})$$

Setting a reset 0 on A has no effect on B.

<u>The completely dephasing channel</u>:
definition : sets all non-diagonal components to 0.

$$\Delta \begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix} = \begin{pmatrix} \alpha_{00} & 0 \\ 0 & \alpha_{11} \end{pmatrix}$$

This transforms a correlated (or intricate) state into a mixed state. (the correlation comes from off-diagonal elements). This process is called “completely dephasing channel”, and represents a well-known process for decoherence. It can also be partially applied. Assuming we have $epsilon \in, (0, 1)$, we define the channel by :

$$\Delta_\epsilon = (1 - \epsilon)Id + \epsilon\Delta$$

which applies the following transformation:

$$\Delta_\epsilon(\rho) = (1 - \epsilon)\rho + \epsilon\Delta(\rho)$$

The off-diagonal elements of the density matrix are not set to 0, but :

$$\Delta \begin{pmatrix} \langle 0|\rho|0\rangle & \langle 0|\rho|1\rangle \\ \langle 1|\rho|0\rangle & \langle 1|\rho|1\rangle \end{pmatrix} \rightarrow \begin{pmatrix} \langle 0|\rho|0\rangle & (1-\epsilon)\langle 0|\rho|1\rangle \\ (1-\epsilon)\langle 1|\rho|0\rangle & \langle 1|\rho|1\rangle \end{pmatrix}$$

<u>The completely depolarizing channel</u>:
This channel always outputs the complete mixed state.

$$\Omega(\rho) = Tr(\rho)\frac{\mathbb{I}}{2}$$

This channel represents an extreme form of noise. It can be applied in the same partial way as the previous channel and we have :

$$\Omega\epsilon(\rho) = (1-\epsilon)\rho + \frac{\epsilon}{2}\mathbb{I}$$

2. Channel representations

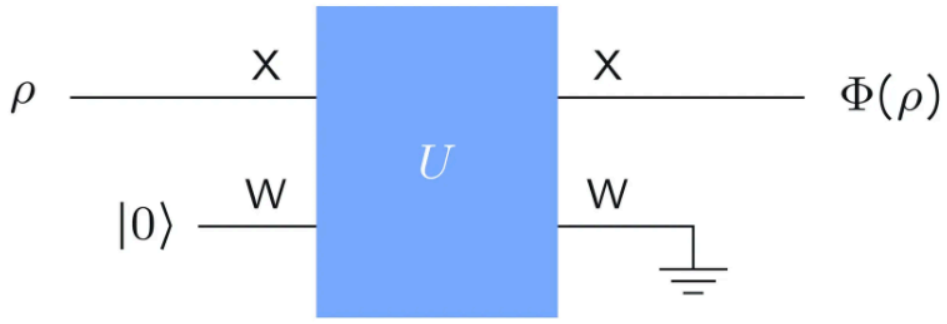
How can we represent an arbitrary channel in an arbitrary way? Remember, linear mapping is matrix-to-matrix, not vector-to-vector.

Stinespring representation

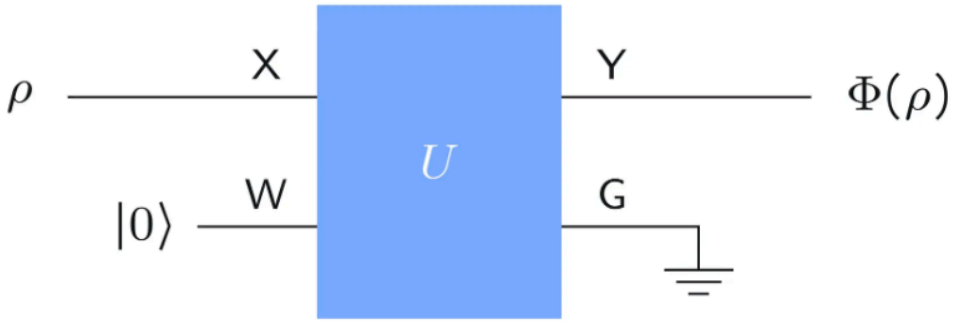
Any channel can be represented as follows:

1. Forms a compound system including an input system and a workspace system initialization.
2. Performs a unit operation on the compound system.
3. Discard all but the output system.

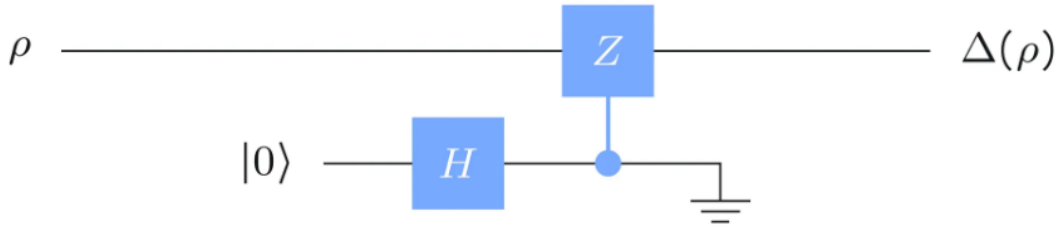
Example for the case of Φ on X



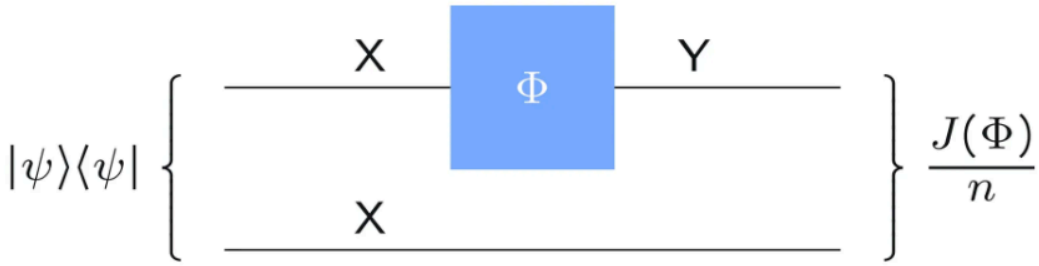
The mass symbol represents the fact that W is discarded. Inputs/outputs can be different



Such a representation is a stinespring channel representation.
Example of an implementation of a complete stinespring channel:



Example: completely dephasing channel (other example)



Kraus representation

The Kraus representation is a convenient formula for expressing the action of the channel on matrix addition and multiplication. In general, the Kraus representation of the Φ channel is of the following form:

$$\Phi(\rho) = \sum_{k=0}^{N-1} A_k \rho A_k^\dagger$$

where $A_0, A_1, \ldots, A_{N-1}$ are matrices that all have the same dimensions.

- The columns of $A_0, A_1, \ldots, A_{N-1}$ correspond to the classical states of the input system.
- The rows of $A_0, A_1, \ldots, A_{N-1}$ correspond to the classical states of the output system.

These matrices must satisfy the following condition:

$$\sum_{k=0}^{N-1} A_k \rho A_k^\dagger = \mathbb{I}$$

Example: qubit reset channel

$$\sum_{k=0}^{1} A_k \rho A_k^\dagger = |0\rangle\langle 0|\rho|0\rangle\langle 0| + |0\rangle\langle 1|\rho|1\rangle\langle 0|$$

$$= (\langle 0|\rho|0\rangle + \langle 1|\rho|1\rangle)|0\rangle\langle 0|$$

$$= Tr(\rho)|0\rangle\langle 0|$$

We can check that the condition is true:

$$\sum_{k=0}^{1} A_k A_k^\dagger = |0\rangle\langle 0||0\rangle\langle 0| + |1\rangle\langle 0||0\rangle\langle 1|$$

$$= |0\rangle\langle 0| + |1\rangle\langle 1| = \mathbb{I}$$

Choi representation

The Choi representation of a Φ channel is a matrix denoted $J(\Phi)$. If the input system has n classical states, and the output system has m classical states, then $J(\Phi)$ is a matrix of dimension $(nm) \times (nm)$.

Key properties of the Choi representation:

> 1. The Choi representation is a faithful representation: for two channels Φ and Ψ we have $J(\Phi) = J(\Psi)$ if $\Phi = \Psi$.
> 2. Simple condition for verifying $J(\Phi)$ is true if Φ is a valid channel.

Note: The matrix $J(\Phi)$ does not directly represent Φ as a linear mapping. However, the action of Φ can be obtained from $J(\Phi)$ by a simple formula.

Suppose we have a channel Φ from a system of X to Y, and we assume we have a classical state of the input system of X in Σ. The Choi representation of Φ is defined as follows:

$$J(\Phi) = \sum_{a,b\in\Sigma} |a\rangle\langle b| \otimes \Phi(|a\rangle\langle b|)$$

If we assume that $\Sigma = \{0, \ldots, n-1\}$, then we can express $J(\Phi)$ as a block matrix:

$$J(\Phi) = \begin{pmatrix} \Phi(|0\rangle\langle 0|) & \Phi(|0\rangle\langle 1|) & \ldots & \Phi(|0\rangle\langle n-1|) \\ \Phi(|1\rangle\langle 0|) & \Phi(|1\rangle\langle 1|) & \ldots & \Phi(|1\rangle\langle n-1|) \\ \vdots & \vdots & \vdots & \vdots \\ \Phi(|n-1\rangle\langle 0|) & \Phi(|n-1\rangle\langle 1|) & \ldots & \Phi(|n-1\rangle\langle n-1|) \end{pmatrix}$$

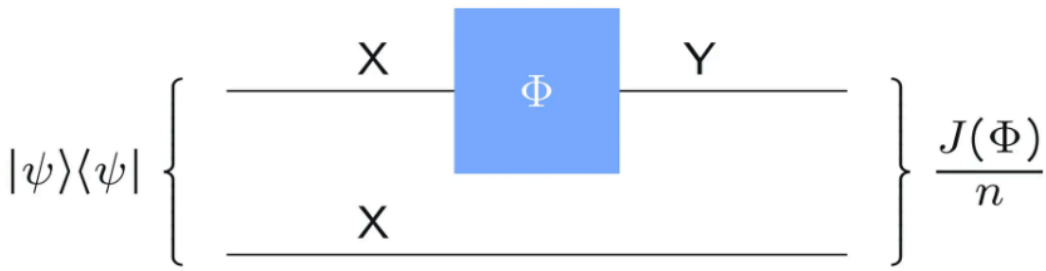
It is a block matrix of the Choi representation, each block $\Phi(|a\rangle\langle b|)$ for each pair (a, b) of an input class state. Note that $|a\rangle\langle b| \quad \forall (a,b) \in [0, n-1[$ forms a basis of the space for any $n \times n$ complex matrix. Don't forget the linearity property of the operation.

If we normalize the Choi representation $J(\Phi)$ of the Φ channel by dividing by $n = |\Sigma|$, we obtain a density matrix. This state is called the Choi state of Φ.

$$|\phi\rangle = \frac{1}{\sqrt{n}} \sum_{a\in\Sigma} |a\rangle \otimes |a\rangle$$

$$|\phi\rangle\langle\phi| = \frac{1}{n} \sum_{a,b\in\Sigma} |a\rangle\langle b| \otimes |a\rangle\langle b|$$

$$\frac{1}{n}J(\Phi) = \frac{1}{n}\sum_{a,b\in\Sigma} |a\rangle\langle b| \otimes \Phi(|a\rangle\langle b|)$$



Implications:

> 1. $J(\Phi)/n$ is a density matrix and therefore $J(\Phi) \geq 0$
> 2. Tracing the system Y from $J(\Phi)/n$ leaves the state of X in a completely mixed state

$$Tr_Y(\frac{J(\Phi)}{n}) = \frac{\mathbb{I}_X}{n} \quad \rightarrow \quad Tr_Y(J(\Phi)) = \mathbb{I}_X$$

These conditions on $J(\Phi)$ are necessary and sufficient for a Φ channel.

Examples:
representation of the “completely dephasing channel”:

$$J(\Delta) = \sum_{a,b=0}^{1} |a\rangle\langle b| \otimes \Delta(|a\rangle\langle b|) = \sum_{a=0}^{1} |a\rangle\langle a| \otimes |a\rangle\langle a| = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Representation of “the completely depolarizing channel”:

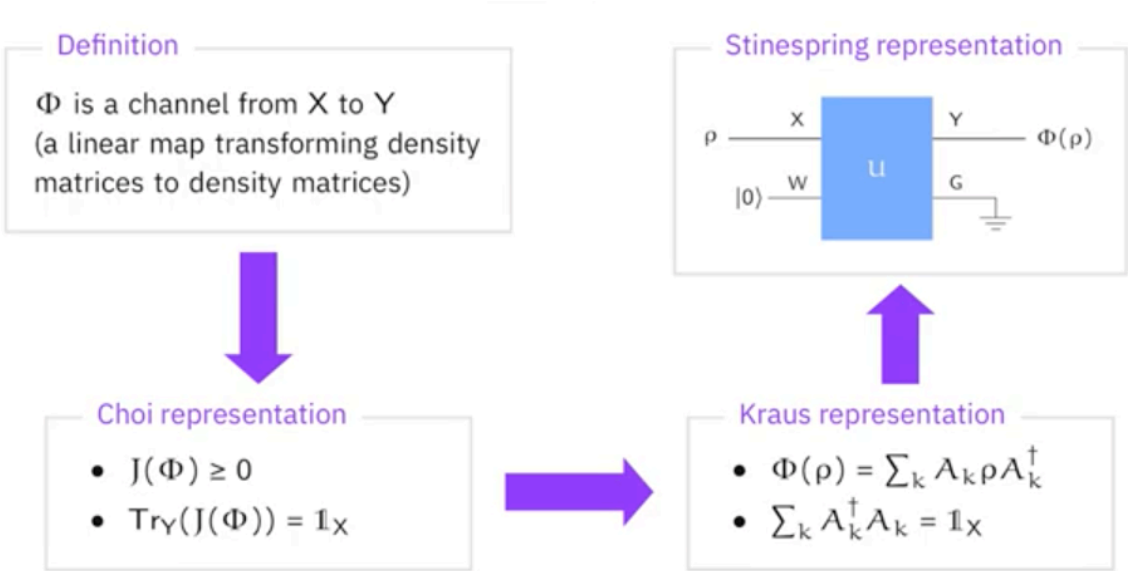
$$J(\Omega) = \sum_{a,b=0}^{1} |a\rangle\langle b| \otimes \Omega(|a\rangle\langle b|) = \sum_{a=0}^{1} |a\rangle\langle a| \otimes \frac{\mathbb{I}}{2} = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}$$

Representation of the “qubit reset channel”:

$$J(\Lambda) = \sum_{a,b=0}^{1} |a\rangle\langle b| \otimes \Lambda(|a\rangle\langle b|) = \sum_{a=0}^{1} |a\rangle\langle a| \otimes |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Representation of “the identity channel”:

$$J(Id) = \sum_{a,b=0}^{1} |a\rangle\langle b| \otimes Id(|a\rangle\langle b|) = \sum_{a,b=0}^{1} |a\rangle\langle b| \otimes |0\rangle\langle b| = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

Note that $J(Id)$ is not the identity matrix. Choi's representation does not directly represent the channel action, which is usually represented by a matrix that represents the linear mapping.

3. Equivalence of representations